

Phase 2: Requirement Analysis Phase

Project Name: **FitSense (FitTrack AI)** – Personalized Fitness Recommendations Powered by AI

2.1 Phase Overview & Analytical Scope

The Requirement Analysis Phase translates the foundational goals established during ideation into rigorous, verifiable software specifications. It details the operational expectations for authentication, workout management, and Google Gemini AI integrations, setting baseline parameters for data contracts, security rules, environment compatibility, and hardware sizing.

2.2 Functional Requirements Specification (FRS)

Module	Requirement Title	Functional Description & Verification Rule	Security / Trigger
Auth	User Registration	Accept name, email, and password. Validate email syntax via regex, verify uniqueness in MongoDB, and hash password using bcrypt (10 salts).	Public Route POST /api/auth/register
Auth	User Login	Validate submitted email and plaintext password against stored bcrypt hash; generate and return a signed JSON Web Token (JWT).	Public Route POST /api/auth/login
Auth	User Profile Retrieval	Extract and verify Bearer token from request header; locate user document by decoded ID; return sanitized user object.	Protected Route GET /api/auth/profile
Workout	Create Workout Record	Validate workoutName, category (enum), duration (>= 1 min), and caloriesBurned (>= 0); link record to authenticated user ObjectId.	Protected Route POST /api/workouts
Workout	Retrieve Workouts	Query all workouts belonging to the calling user; allow single workout retrieval via MongoDB _id parameter.	Protected Route GET /api/workouts/:id
Workout	Update / Delete Record	Permit modification of workout fields with schema re-validation; support hard removal of records belonging to the authenticated user.	Protected Route PUT/DELETE /workouts/:id
Workout	Multi-Field Search	Filter personal workout histories using query criteria matching workout name, exercise category, or specific workout date.	Protected Route GET /api/workouts/search
AI Module	Workout Plan Synthesis	Accept user age, fitnessGoal, and experience level; query Google Gemini API; return structured weekly schedule, exercises, and safety tips.	Public/User Route POST /api/ai/workout-rec
AI Module	Fitness Insights	Accept totalWorkouts, averageDuration, and totalCaloriesBurned; analyze cumulative metrics; return evaluation and progress suggestions.	Public/User Route POST /api/ai/fitness-insights

2.3 User Requirements

- Autonomous Account Setup:** Users must be able to register using name, email, and password.
- Private Record Management:** Users must only access

2.4 System & Non-Functional Requirements

- Cryptographic Security:** Zero plaintext passwords stored; JWT token validation on protected endpoints.
- Data Schema Integrity:** Strict boundary assertions on

2.5 Hardware & Software Environment Requirements

Hardware Requirements Specification

Hardware Component	Documented Specification
Processor	Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent
System Memory (RAM)	8 GB Minimum (16 GB Recommended for concurrent runtime, DB & nodes)
Disk Storage	1 GB of available disk workspace
Network Interface	Standard TCP/IP connectivity for outbound Google Gemini AI API calls

Software Requirements Specification

Software Layer	Documented Specification
Operating System	Windows 10/11, macOS, or Linux (cross-platform)
Server Runtime	Node.js (v16 or above)
Package Manager	npm (v8 or above)
Web Framework	Express.js (RESTful routing engine)
Database Engine	MongoDB (NoSQL Document Store)
Verification Tool	Postman / Thunder Client (VS Code)

2.6 Input & Output Data Requirements

Transaction Name	Required Input Fields (Payload)	Expected Output Data (JSON Structure)
User Registration	name (String), email (String), password (String >= 6 chars)	success: true, token (JWT String), user: { id, name, email }
Workout Entry	workoutName, category, duration (Mins), caloriesBurned, workoutDate	success: true, data: { _id, user, workoutName, category, duration, caloriesBurned, workoutDate, timestamps }
AI Workout Plan	age (Number), fitnessGoal (String), experience (String)	recommendation (String text detailing split, exercises, sets, safety advice)
AI Fitness Insights	totalWorkouts (Number), averageDuration (Number), totalCaloriesBurned (Number)	insight (String text analyzing volume, consistency, progressive overload)

2.7 Technologies & Frameworks Documented

Node.js (v16+)

Express.js

MongoDB

Mongoose ODM

bcryptjs (v10)

jsonwebtoken (JWT)

cors

dotenv

nodemon (dev)

Google Gemini AI

Thunder Client / Postman

2.8 Constraints and Operational Assumptions

Documented Constraints: (1) The application is developed strictly as a backend REST service. (2) Exercise categories are limited to supported enum entries: Cardio, Strength Training, Yoga, Running, Cycling, Walking. (3) duration must be >= 1 minute; caloriesBurned must be >= 0. (4) Authentication middleware strictly enforces standard Authorization: Bearer <token> headers.

Operational Assumptions: (1) Continuous network connectivity for Google Gemini API resolution. (2) Front-end clients securely persist issued JWT tokens across sessions. (3) MongoDB instance maintains available socket connections on local port 27017 or remote cluster.